

Drive for better vision



OTA User Guide V1.2

Himax Technologies, Inc
奇景光電股份有限公司

Agenda

- **Check Before Starting OTA**
- **OTA Upgrade Operations**

Himax Technologies, Inc.
Confidential
Do Not Copy

Check Before Starting OTA

Check Before Starting OTA

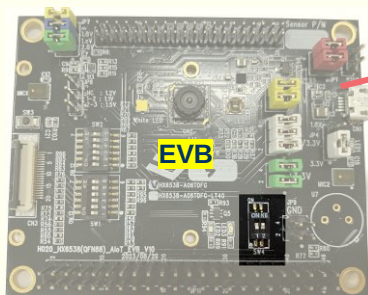
1. Does the EVB already been burned with any firmware?

- ❖ If do, the EVB can upgrade firmware through OTA.
- ❖ If not, the EVB **CANNOT** upgrade firmware through OTA.
To use OTA, there must be any firmware on the EVB so that it can boot up correctly.

To learn more about burning firmware with PC tool,
please refer to “[2_EVK_and_PCTool_User_Guide_HX6538_AIoT_EVB_QFN88](#)”.

2. Do the EVB and debug board already set to the normal mode?

- ❖ If not, two steps to set normal mode:
 - ① Turn off the EVB power.
 - ② Set the switches on EVB: **SWITCH [1, 2] = [OFF, OFF]**



SWITCH SETTING	MODE
 1 2 [OFF, OFF]	EVB: Normal mode

OTA Upgrade Operations

OTA Upgrade Operations (1/6)

- Turn on the power of EVB
- Launch OTA tool
(path: SDK_root/we2_ota_tool/ota.exe)
- Operate by following steps:

① Key in “1” (1. I2C protocol)

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Please choice transfer protocol:
1.I2C protocol
2.UART protocol
0.exit
1
```

② Key in “1” (1. Command)

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : I2C
=====OTA Main Menu=====
Please choice following one item:
1.Command
2.Upgrade
3.Others
0.exit
1
```

OTA Upgrade Operations (2/6)

③ Key in “8” (8. Intoupg)

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : I2C
=====OTA Command Menu=====
Please choice following one item:
1.Get OTA Loader Version.....[OTA LOADER]
2.Read WE2 Chip id.....[OTA LOADER]
3.Getstatus (Get WE-II current status).....[OTA LOADER]
4.ReBoot WEII.....[OTA LOADER]
5.Busy pin select.....[OTA LOADER]
6.Read busy pin select.....[OTA LOADER]
7.Reserved.....[NONE]
8.Intoupg.....[APPLICATION]
0.exit/back
  8
```

Hint: If you also start a log monitor, the console of log will show that:

```
[MSG] feature_type: 0x50
[MSG] cmd_id: 0x8
[MSG] payload_len: 0x0
hx_lib_i2ccomm_validate_checksum
Payload len:0x0000 CRC_val: 0x10f5, CRC_cal:0x10f5
===I2CCOMM_CMD_OTA_JUMP2UPG===
1st BL Build DATE=Aug  1 2023, 0x00010001
jump_addr=0x3401f000
Compiler Version: ARM CLANG, Clang 12.0.0 (ssh://ds-gerrit/armcompiler/llvm-project e64d644232aba72041b013571dc92e0d3fb7b4e2)
set_IP_secure done
flash type[2], flash size[5]
2ndBootLoader Version : SVN_183742, Feb  6 2024, 14:17:28
Chip Version : 0x8538000c
OTA I/F : I2C UART
SecureLib Version : SVN_181742
Serial number : 0x0
Part number : 0x0
[WE2 OTA FIRMWARE(Feb  6 2024, 14:17:28)] Daemon Process
Event Handler
evt_uartcomm_init UART1 Baudrate = 115200
```

OTA Upgrade Operations (3/6)

④ Press any key

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : I2C
=====OTA Command Menu=====
Please choice following one item:
1.Get OTA Loader Version.....[OTA LOADER]
2.Read WE2 Chip id.....[OTA LOADER]
3.Getstatus (Get WE-II current status).....[OTA LOADER]
4.ReBoot WEII.....[OTA LOADER]
5.Busy pin select.....[OTA LOADER]
6.Read busy pin select.....[OTA LOADER]
7.Reserved.....[NONE]
8.Intoupg.....[APPLICATION]
0.exit/back

                               8
Intoupg
Press Enter key to continue...
```

⑤ Key in “0” (0. exit/back)

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : I2C
=====OTA Command Menu=====
Please choice following one item:
1.Get OTA Loader Version.....[OTA LOADER]
2.Read WE2 Chip id.....[OTA LOADER]
3.Getstatus (Get WE-II current status).....[OTA LOADER]
4.ReBoot WEII.....[OTA LOADER]
5.Busy pin select.....[OTA LOADER]
6.Read busy pin select.....[OTA LOADER]
7.Reserved.....[NONE]
8.Intoupg.....[APPLICATION]
0.exit/back

                               0
```


OTA Upgrade Operations (4/6)

⑥ Key in “2” (2. Upgrade)

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : I2C
=====OTA Main Menu=====
Please choice following one item:
1.Command
2.Upgrade
3.Others
0.exit
```

2

⑦ Key in “1” (1. UPGRADE ALL IMAGE(Upgrade all image file))

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : I2C
=====OTA Upgrade Menu=====
Please choice following one item:
1.UPGRADE ALL IMAGE(Upgrade all image file)
2.UPGRADE All APPLICATIONs
3.UPGRADE NS APPLICATIONs
0.exit/back
```

1

OTA Upgrade Operations (5/6)

⑧ Key in “1” (1. output.img)

```
Upgrade Image
=====File List=====
Please select file:
1. output.img
0. exit
=====File List=====
1
```

⑨ OTA upgrade started ...

```
img_file = img\output.img
cstm_layout_file = img\cstm_layout.bin
Getstatus
status = 0x000c0000

==>OTA_STATUS_OTA_LOADER_IDLE
status = 0x000c0000

==>OTA_STATUS_OTA_LOADER_IDLE
hx_img_upgrade Upgrade Started
hx_memdsp->info.num_sbs = 0x6
part_idx = 0 pat_type = 0
start cmd : 0x0, 0x4ac0001, 0x15000, 0x14000, 0xbaa0
-count_length = 0x0
Getstatus
status = 0x00130000

==>OTA_STATUS_OTA_LOADER_INIT
status = 0x00130000

==>OTA_STATUS_OTA_LOADER_INIT
-count_length = 0x0
\count_length = 0x100
/count_length = 0x200
-count_length = 0x300
\count_length = 0x400
```

OTA Upgrade Operations (6/6)

⑩ OTA upgrade finished

```
==>OTA_STATUS_OTA_RECEIVE_UPGRADE_DATA_DONE
.
Burn flash successs
Getstatus
status = 0x000c0000

==>OTA_STATUS_OTA_LOADER_IDLE
status = 0x000c0000

==>OTA_STATUS_OTA_LOADER_IDLE
ReBoot WE2
upgrade success Upgrade Finished
Press Enter key to continue...
```